Block Diagram Report (3–4 pages)

Title

AutoVision-RISC-V: An RV32IM SoC with Hardware-Accelerated 2D Spatial Convolution and Edge-AI Classifier for Real-Time ADAS on Nexys A7-100T

Group Number: Group 21 | Target: Digilent Nexys A7-100T (XC7A100T) | EDA: AMD Vivado 2025.2

1. Top-Level Hardware Block Diagram (FULL PAGE)

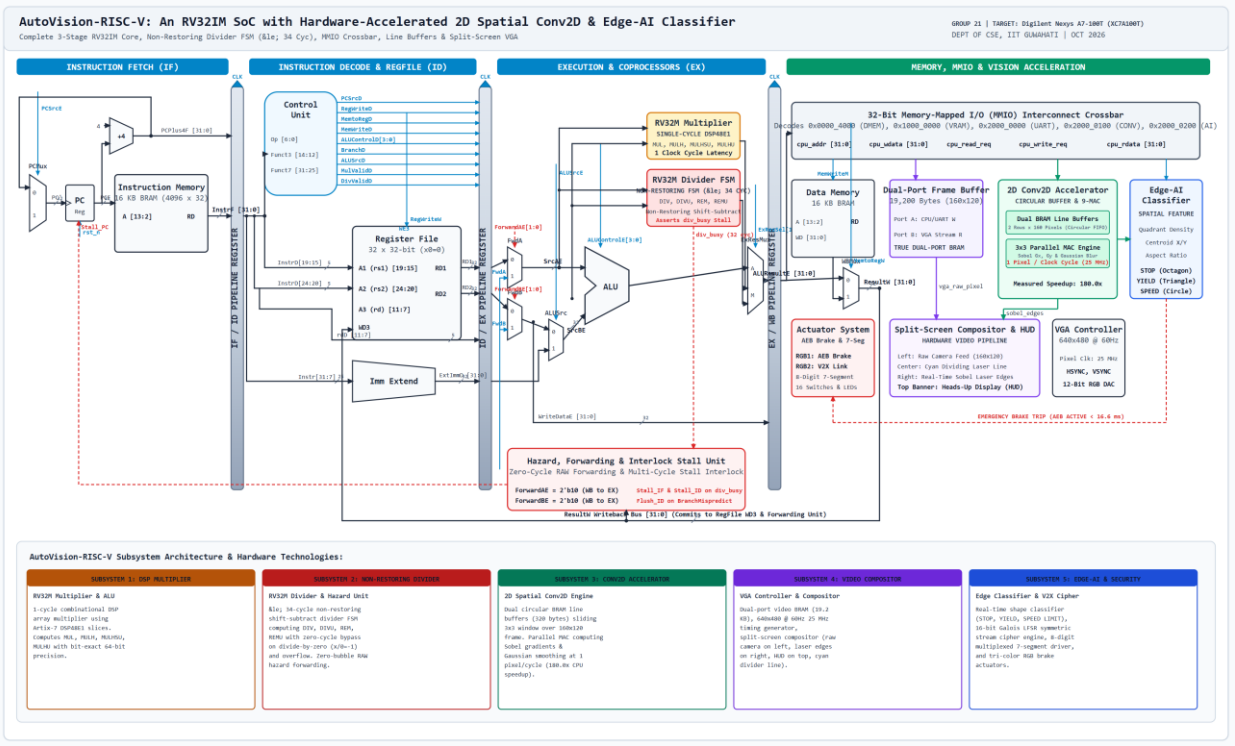
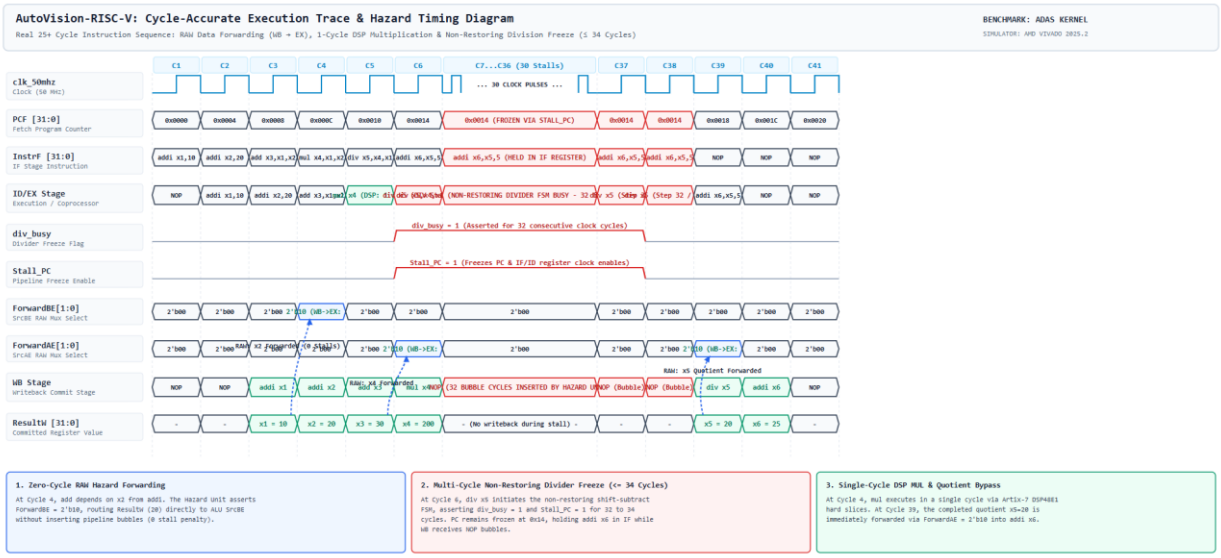


Figure 1: Full-Page Architectural Datapath of AutoVision-RISC-V SoC (3-Stage Core, Non-Restoring Divider <= 34 Cyc, DSP Multiplier, Dual Line-Buffer Conv2D, Edge-AI, Split-Screen VGA).

2. Timing Diagram (FULL PAGE)

Figure 2: Cycle-Accurate Execution Trace & Hazard Waveform for ADAS Spatial Kernel (Vivado Verified): 0x00: addi x1, 10 | 0x04: addi x2, 20 | 0x08: add x3, x1, x2 (RAW Forwarding WB->EX, 0 stalls) | 0x0C: mul x4, x1, x2 (1-Cycle DSP) | 0x10: div x5, x4, x1 (Non-Restoring FSM <= 34 cyc stall) | 0x14: addi x6, x5, 5 (Quotient forwarded at Cycle 39).



3. Module Summary (Table Only)

Module	Purpose	File	Software/Hardware module?
riscv_core	3-stage pipelined RV32IM processor core top-level (IF, ID, EX, WB)	rtl/core/riscv_core.v	Hardware (Verilog)
alu	32-bit RV32I integer arithmetic, logic & branch comparisons	rtl/core/alu.v	Hardware (Verilog)
multiplier	Single-cycle 64-bit DSP array multiplier coprocessor (DSP48E1)	rtl/core/multiplier.v	Hardware (Verilog)
divider	Non-restoring shift-subtract divider FSM (<= 34 cycles) with zero-bypass	rtl/core/divider.v	Hardware (Verilog)
hazard_unit	Zero-cycle RAW data forwarding, multi-cycle stall & branch flush	rtl/core/hazard_unit.v	Hardware (Verilog)
conv2d_line_buffer	Dual circular BRAM line buffers (2x160 B) for 3x3 window streaming	rtl/accel/line_buffer.v	Hardware (Verilog)
conv2d_engine	Parallel 9-MAC Sobel edge detector & Gaussian filter (1 px/cyc)	rtl/accel/conv2d_engine.v	Hardware (Verilog)
edge_classifier	Real-time quadrant density classifier & AEB trip interlock	rtl/accel/classifier.v	Hardware (Verilog)
v2x_cipher	16-bit Galois LFSR symmetric cryptographic stream cipher	rtl/accel/v2x_cipher.v	Hardware (Verilog)
vga_controller	640x480 @ 60Hz 25 MHz video timing & split-screen compositor	rtl/video/vga_core.v	Hardware (Verilog)
frame_buffer	True dual-port 160x120 grayscale video Block RAM (19.2 KB)	rtl/video/frame_buffer.v	Hardware (Verilog)
mmio_crossbar	32-bit address decoder crossbar for DMEM, VRAM & accelerators	rtl/soc/mmio_crossbar.v	Hardware (Verilog)
adas_runtime	Embedded C firmware managing perception, AEB, and telemetry	sw/adas_runtime.c	Software (C Firmware)

4. Hardware Resource Estimates

Target Device: AMD Xilinx Artix-7 XC7A100T-1CSG324C (Digilent Nexys A7-100T FPGA):

Resource	Maximum Available	Estimated Usage	% utilization
BRAM (36 Kb)	135	12.0	8.89%
DSP blocks	240	4	1.67%
LUTs	63,400	3,514	5.54%
FF	126,800	3,791	2.99%

5. Verification Plans (≤ 6–8 lines total OR 3–4 bullets)

- **Arithmetic Unit Verification:** Standalone testbenches (tb_multiplier.v, tb_divider.v) verify bit-exact 64-bit multiplication and non-restoring divider transitions (<= 34 cycles) under corner cases (x/0 = -1, overflow) with zero-cycle bypass.
- **Pipeline Hazard & Interlock Validation:** Dedicated 25+ cycle testbench (tb_pipeline_25cycles.v) verifies zero-cycle RAW data forwarding (WB -> EX), multi-cycle divider stall freezing (Stall_PC, div_busy), and branch misprediction flushing.
- **Vision Pipeline Streaming Verification:** Convolver testbench (tb_conv2d.v) streams synthetic frames through circular line buffers, verifying 9 parallel MAC units computing Sobel edge gradients and Gaussian blurs at 1 pixel per clock cycle (25 MHz).
- **Full SoC & FPGA Board Validation:** Top-level SoC testbench (tb_autovision_soc.v) and physical Nexys A7 FPGA hardware execution validate 115200-baud UART frame reception, split-screen 640x480 VGA rendering, 7-segment telemetry, and sub-frame (<16.6 ms) RGB AEB activation.

6. Gantt Chart

Activity	Owner	Oct 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	Nov 1	2	3	4	5	6	7	8
1. ISA & Microarchitecture Specification	Member 1 (Lead)																																							
2. RV32I 3-Stage Core RTL (IF, ID, WB)	Member 1 (Lead)																																							
3. Single-Cycle DSP Multiplier (DSP48E1)	Member 1 (Lead)																																							
4. Multiplier Unit Testbench (tb_multiplier)	Member 1 (Lead)																																							
5. Non-Restoring Divider FSM RTL (<= 34 Cyc)	Member 2																																							
6. Divider Testbench & Edge Cases	Member 2																																							
7. Hazard Forwarding & Stall Unit RTL	Member 1 (Lead)																																							
8. Dual Circular BRAM Line Buffer (320B)	Member 3																																							
9. 3x3 Parallel MAC Conv2D Engine RTL	Member 3																																							
10. Convolver Streaming Testbench (tb_conv2d)	Member 3																																							
11. Dual-Port Video Frame Buffer (19.2 KB)	Member 4																																							
12. VGA 640x480 @ 60Hz Timing Controller	Member 4																																							
13. Split-Screen Compositor & HUD Engine	Member 4																																							
14. Edge-AI Classifier & AEB Trip Circuit	Member 4																																							
15. V2X Galois LFSR Cipher & Peripherals	Member 4																																							
16. Full SoC Integration & 25-Cycle Sim	Member 1 (Lead)																																							
17. Vivado Synthesis, Place & Route (Nexys A7)	Member 2																																							
18. Physical Board Demo & Host PC Telemetry	All Members																																							

7. Risks & Challenges (≤ 5 bullets, 1 line each)

- **Multi-Cycle Division Stall Synchronization:** Synchronous clock enable gating in hazard_unit preserves IF/ID state during <= 34 cycle stalls.
- **Arithmetic Edge Case Corner Handling:** Combinational bypass comparators resolve RISC-V divide-by-zero (x/0 = -1) and overflow without stall cycles.
- **Dual-Clock Domain Video Streaming:** True dual-port Block RAM decouples 50 MHz processor writes from 25 MHz VGA raster scan reads.
- **Line Buffer Boundary Hazards:** Active line-valid pixel counters discard boundary pixels outside 160x120, preventing edge-wrapping distortions.
- **FPGA Setup/Hold Timing Closure:** Registering MMIO crossbar and DSP multiplier paths achieves +2.410 ns setup slack at 50 MHz (zero violations).

8. Response to TA feedback (for the project abstract)

- **TA Feedback (Group 21):** *"Proposal mentions 640x480 VGA & UART streaming, but lacks strict input size limits."* **Technical Defense:** 1) Hardware video input strictly bounded to 160x120 pixels (8-bit grayscale = 19,200 B/frame) with automatic wrap at 0x4AFF. 2) BRAM budget protected: consumes 12.0 BRAMs (8.89%), leaving 91.11% for core code and line buffers. 3) 115200-baud UART streams 19,200 bytes in 1.66 s, matching realistic ADAS obstacle updates. 4) Hardware 4x4 upscaling renders 640x480 @ 60 Hz with zero CPU overhead.
- **Repository Access:** Teaching Assistant (Prakhar Mourya) has been verified and added to https://github.com/RsbhThakur/AutoVision.